Boshi Wang

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Education

ShanghaiTech University Shanghai, China
Major in Computer Science Sep. 2016 - Present

- GPA: 3.76/4.0
- Awards: Dean's Scholarship (2016), Excellence Scholarship (2016-2017), Dean's Scholarship (2017-2018)
- Selected Coursework: Applied Algebraic Geometry, Convex Optimization, Matrix Analysis, Introduction to Algorithmic Game Theory, Subspace Learning (Fall 2019)

Publication

* denotes equal contribution

Published

- Z. Tu, **B. Wang**, T. Duan, and F. Gao. Pattern-learning-based Noise Elimination Algorithm in Photoacoustic Sensing and Imaging. In *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'18)*, Honolulu, HI, USA, on July 17-21, 2018.

Under Review

- Y. Guo*, **B. Wang***, D. Hughes, K. Sycara. Designing Context-Sensitive Norm Inverse Reinforcement Learning Framework for Norm-Compliant Autonomous Agents. *Submitted.*

In Progress

- **B. Wang**, M. Tsakiris. Homomorphic Sensing with Union of Linear Subspaces as Input Space. *Link*: https://boshi-wang.github.io/documents/HS_with_Union_of_Subspaces.pdf

Research Experience

Research Interest: **Machine Learning, Artificial Intelligence, Robotics**

Advanced Agent-Robotics Technology Lab, Carnegie Mellon University Pittsburgh, USA
Undergraduate Research Intern, supervised by Prof. *Katia Sycara* Jun 2019 - Sep 2019

- Developed Context-Sensitive Norm Inverse Reinforcement Learning (CNIRL), a computational scalable IRL framework for developing norm-aware autonomous agents through learning from demonstrations under large number of environmental contexts.

Manolis Research Group, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. *Manolis C. Tsakiris* Feb 2019 - Present

- Extended Homomorphic Sensing (a recent ML research topic proposed by Prof. Tsakiris in an *ICML'19* paper) into the case where the input space is a union of finitely many linear subspaces instead of a single linear subspace.

- Currently conducting theoretical analysis on solving subspace recovery under partially permuted data with Dual Principle Component Pursuit (DPCP), and also on further extending Homomorphic Sensing into the case where the input space is an algebraic variety defined by arbitrary generic homogeneous polynomials.

Hybrid Imaging System Laboratory, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. Fei Gao Jan 2018 - Nov 2018

- Developed an algorithm that can learn from historical data about the pattern of a periodic noise in Photoacoustic signals and efficiently perform noise reduction.
- Conducted research on using denoising autoencoders (DAE) for photoacoustic signal noise reduction. The denoising behavior largely outperforms traditional methods under low signal-to-noise ratio.

Mobile Autonomous Robotic Systems Lab, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. *Sören Schwertfeger* Jun 2017 - Dec 2017

- Implemented an algorithm which can enable a robot to localize itself based completely on visual information of Aruco markers in surrounding environment collected by a camera on the robot. The work contributes to a bigger project in the lab about Simultaneous Localization and Mapping.

Teaching Experience

Teaching Assistant of *SI140: Probability and Mathematical Statistics*, ShanghaiTech University.
Course Instructor: Prof. *Ziyu Shao* Shanghai, China

Internship

Chinese Software Developer Network Corporation (CSDN) Beijing, China
Intern of System Developer Jun 2018 - Sep 2018

- Optimized the keyword matching and filtering algorithm of the blog recommendation system.
- Developed an efficient method for clustering search queries of blogs and implemented the method with parallel computing using Spark.

Other Selected Projects

Deep Q-learning based AI agent for Beating the Landlord
Course project of *CS181: Artificial Intelligence* in ShanghaiTech University, Fall 2018

- Developed a Deep Q-learning framework which learns to play a popular 3-player card game in China called “Beating the Landlord” through three agents’ self-play.
- Although the trained agent still can’t beat human players, it largely outperforms greedy agents which are used in most online game platforms.

Sparse Representation Based Face Recognition
Course project of *SI131: Linear algebra* in ShanghaiTech University, Fall 2017

- Implemented a face recognition algorithm based on Principle Component Analysis and Sparse Representation. The algorithm achieves 91.4% accuracy on Extended Yale Face Database B.

Miscellaneous

Programming Skills: Python, C/C++, Matlab, Rust, MIPS assembly

Tools: CVX, Tensorflow, Pytorch, Robot Operating System (ROS)

Sports: Sprint (100 meters in 6.2 seconds), Standing Long Jump (2.9 meters), Soccer, Billiard

Other Interests: Guitar, Dota2, a big fan of Japanese animation *Ghost in the Shell*.